

Summary

This video explains the foundational Theory of Computation concepts related to substrings, prefixes, suffixes, and their combinatorial properties. It establishes formal definitions, derives counting formulas, and analyzes the effect of symbol repetition on substring counts.

Key conclusions:

- The number of substrings of a string of length n ranges from $n + 1$ to $(n(n + 1))/2 + 1$.
- Prefix and suffix counts remain constant ($n + 1$), independent of symbol repetition.
- Repetition reduces the number of distinct substrings but does not affect positional constructs such as prefixes and suffixes.
- Advanced combinatorial analysis allows calculation of strings containing at least one pair of consecutive identical symbols.

1. Fundamental Definitions of String Components

1.1 Substrings

A substring is a contiguous sequence of symbols within a string. If u is a substring of W , then u must appear in W without reordering or skipping symbols.

Contiguity Requirement:

For $W = \text{"TOC"}:$

- "TO" is a substring.
- "TC" is not a substring (symbols are not contiguous).

Length Constraint:

If u is a substring of W , then $|u| \leq |W|$.

Universal Properties:

- Every string is a substring of itself.
- The empty string (ϵ) is a substring of every string.

1.2 Prefixes and Suffixes

Prefix:

A substring starting from the first symbol.

Example for "TOC":

ϵ , T, TO, TOC

Proper Prefix:

All prefixes except the string itself.

Example:

ϵ , T, TO

Suffix:

A substring ending at the final symbol.

Example for "TOC":

ϵ , C, OC, TOC

Proper Suffix:

All suffixes except the string itself.

Example:

ϵ , C, OC

2. Categorization of Substrings

Substrings are divided into trivial and non-trivial categories.

Category	Description	Included Elements
Trivial Substrings	Present in every string	ϵ and the string itself (W)
Non-Trivial Substrings	All other substrings	All u such that $u \neq W$ and $u \neq \epsilon$

Key Insight:

The number of trivial substrings is always exactly 2.

3. Mathematical Analysis and Formulas

3.1 Distinct Symbol Case

If a string of length n consists entirely of distinct symbols:

Metric	Formula
Total Substrings	$(n(n + 1))/2 + 1$
Non-Empty Substrings	$(n(n + 1))/2$
Non-Trivial Substrings	$(n(n + 1))/2 - 1$

Number of Prefixes	$n + 1$
Number of Suffixes	$n + 1$
Proper Prefixes/Suffixes	n

Explanation:

For each starting index, there are progressively fewer ending indices, producing the triangular number formula.

3.2 Impact of Symbol Repetition

Substrings:

Symbol repetition reduces the number of distinct substrings.

Minimum Case:

If all symbols are identical (e.g., "AAAAA"):

Total substrings = $n + 1$

Range:

For any string of length n :

$n + 1 \leq \text{Number of Substrings} \leq (n(n + 1))/2 + 1$

Prefixes and Suffixes:

Remain constant at $n + 1$ regardless of repetition, since they depend on positions, not distinctness.

4. Relationships Between String Components
5. Prefix-Suffix Symmetry:
Number of prefixes = Number of suffixes = $n + 1$.
6. Inclusion Principle:
Every prefix and every suffix is a substring.
7. Exclusion Principle:
Not every substring is a prefix or suffix.

Example:

In "TOC", substring "O" is neither prefix nor suffix.

4. Commonality:
 ϵ and the string itself belong to:
 - Prefix set
 - Suffix set
 - Substring set
5. Advanced Combinatorial Analysis: Consecutive Symbols

A common TOC problem involves counting strings with at least one pair of consecutive identical symbols.

5.1 Case Study: Length 5 Ternary Alphabet

Alphabet:

$$\Sigma = \{a, b, c\}$$

Length:

$$n = 5$$

Total possible strings:

$$|\Sigma|^n = 3^5 = 243$$

Strings with NO consecutive identical symbols:

- Position 1: 3 choices.
- Positions 2–5: 2 choices each (cannot match previous symbol).

Total:

$$3 \times 2^4 = 3 \times 16 = 48$$

Strings with at least one consecutive identical pair:

$$243 - 48 = 195$$

5.2 Generalized Formula

For alphabet size $k = |\Sigma|$ and string length n :

Total strings:

$$k^n$$

Strings with no consecutive identical symbols:

$$k \times (k - 1)^{(n - 1)}$$

Strings with at least one consecutive identical pair:

$$k^n - [k \times (k - 1)^{(n - 1)}]$$

This formula applies to binary, ternary, quaternary, and general k-ary alphabets.

6. Case Studies and Data Points

String	Length (n)	Total Substrings	Trivial Substrings	Notes
--------	---------------	---------------------	-----------------------	-------

TOC	3	7	2	All symbols distinct
GATE	4	11	2	All symbols distinct
ALGORITHMS	10	56	2	All symbols distinct
MOBILE	6	22	2	All symbols distinct
TOT	3	6	2	Symbol repetition present
TTT	3	4	2	All symbols identical
AAAAA	5	6	2	All symbols identical (n + 1)